13 B Gradient of a-x graph is $-\omega^2$.

P's ω^2 is four times that of Q's. P's amplitude is half of Q's.

Given $E_K = \frac{1}{2}mv_0^2 = \frac{1}{2}m\omega^2 x_0^2$, the maximum kinetic energy (this is also called the energy of oscillation) of P is the same as that of Q's.

14 B